

```
# Inst Rename notebook
```

```
!pip install transformers
```

```
!pip install torch
```

```
# Import
```

```
from transformers import pipeline
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.12.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.3.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.24.4)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (24.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.1)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2024.5.15)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.20.1)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.12.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.4.3)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.66.1)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (2024.5.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.1.2)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.27.0)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub) (0.12.1)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub) (4.10.0)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0) (4.6.2)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0) (2024.7.4)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0) (1.0.5)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0) (3.10)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*) (0.14.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer) (8.1.7)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer) (13.7.1)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer) (0.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich) (2.18.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py) (0.1.2)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.12.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.10.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (68.0.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.3)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.4)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2024.5.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2) (2.1.5)
```

```
import transformers
```

```
print(transformers.__version__)
```

```
5.0.0
```

```
!pip install --upgrade transformers
```

```
!pip install --upgrade torch
```

Rename notebook

```
Downloading nvidia nvtx-13.0.85-py3-none-manylinux1 x86_64.whl.metadata (1.0
```

```

Collecting cuda-pathfinder==1.1 (from cuda-bindings<14,>=13.0.3->torch)
The old cuda binding cuda-pathfinder1.1 is deprecated. Migrating your old cache. This
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from
Require Rename notebook satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from ji
Downloading libtorch2.1.0-cp312-cp312-manylinux_huggingface_0.8.0-0.8.0-py3.12: UserWarning:
The secret `HF_TOKEN` does not exist in your 560.17/550.7 MB 1.6 MB/s eta 0:00:00
Downloading cuda with the Hugging Face PyTorch create a token in your settings tab (https://huggingface.com/settings/tokens)
You will be able to change this secret in all of your notebooks
Downloading nvidia_cudnn_cu13-9.0.56-py3-none-manylinux2014_x86_64.whl (366.1 MB)
Please note that authentication is recommended to access public models or datasets
566.17/366.1 MB 5.0 MB/s eta 0:00:00
Downloading nvidia_cusparselt_cu13-0.8.0-py3-none-manylinux2014_x86_64.whl (169.9 MB)
5.7 MB/s eta 0:00:00
model.safetensors: 100%
5.7 MB/s eta 0:00:00, 56.7 MB/s]
Downloading nvidia_nccl_cu13-2.28.9-py3-none-manylinux_2_18_x86_64.whl (196.5 MB)
generation_config.json: 100%
1803/368a [0:00:00.00, 31.1 kB/s]
Downloading nvidia_nvshmem_cu13-3.4.5-py3-none-manylinux2014_x86_64.whl (60.
vocab.json: 899k/? [00:00:00.00, 13.7 MB/s] 60.4/60.4 MB 11.9 MB/s eta 0:00:00
Downloading triton-3.6.0-cp312-cp312-manylinux_2_27_x86_64.whl (188.3 MB)
merges.txt: 456k/? [00:00:00.00, 21.5 MB/s] 188.3/188.3 MB 7.0 MB/s eta 0:00:00
Downloading nvidia_cublas-13.1.0.3-py3-none-manylinux_2_27_x86_64.whl (423.1 MB)
tokenizer.json: 1.36M/? [00:00:00.00, 27.7 MB/s] 423.1/423.1 MB 2.7 MB/s eta 0:00:00
Downloading nvidia_cuda_nvrtc-13.0.88-py3-none-manylinux2010_x86_64.whl (90.2/90.2 MB 9.8 MB/s eta 0:00:00

```

```

!pip uninstall transformers -y
!pip uninstall tokenizers -y
!pip uninstall huggingface-hub -y

```

```

Found existing installation: transformers 5.3.0/1.2 MB 54.1 MB/s eta 0:00:00
Downloading nvidia_cuda_nvrtc-13.0.88-py3-none-manylinux2010_x86_64.whl (90.2/90.2 MB 9.8 MB/s eta 0:00:00
Successfully uninstalled transformers-5.3.0/1.2 MB 54.1 MB/s eta 0:00:00
Found existing installation: tokenizers 0.22.2/200.9/200.9 MB 6.8 MB/s eta 0:00:00
Uninstalling tokenizers-0.22.2:
200.9/200.9 MB 6.8 MB/s eta 0:00:00
Downloading nvidia_cuda_nvrtc-13.0.88-py3-none-manylinux2010_x86_64.whl (90.2/90.2 MB 9.8 MB/s eta 0:00:00
Found existing installation: huggingface-hub 1.40.7/145.9 MB 7.1 MB/s eta 0:00:00
Downloading nvidia_cuda_nvrtc-13.0.88-py3-none-manylinux2010_x86_64.whl (90.2/90.2 MB 9.8 MB/s eta 0:00:00
Successfully uninstalled huggingface-hub-1.40.7/145.9 MB 7.1 MB/s eta 0:00:00
Downloading nvidia_cuda_nvrtc-13.0.88-py3-none-manylinux2010_x86_64.whl (90.2/90.2 MB 9.8 MB/s eta 0:00:00

```

```

texts = [

    """Artificial Intelligence (AI) is rapidly transforming the modern world. It is used in healthcare t

    """Climate change is one of the most critical global challenges today. Rising temperatures, meltin

    """Machine learning is a subset of artificial intelligence that focuses on building systems that l

    """In recent years, social media has become an essential part of daily communication. Platforms li

    """The history of space exploration has been marked by significant achievements. From the first hu
]

```

```

for text in texts:
    summary = summarizer(text, max_length=80, min_length=30, do_sample=False)

    print("ORIGINAL:\n", text)
    print("\nSUMMARY:\n", summary[0]['summary_text'])
    print("=="*80)

```

```

ORIGINAL:
    Artificial Intelligence (AI) is rapidly transforming the modern world. It is used in healthcare t

SUMMARY:
    Artificial Intelligence (AI) is rapidly transforming the modern world. It is used in healthcare t
=====
ORIGINAL:
    Climate change is one of the most critical global challenges today. Rising temperatures, melting

SUMMARY:
    Climate change is one of the most critical global challenges today. Governments worldwide are try
=====

```

Your max_length is set to 80, but your input_length is only 77. Since this is a summarization task
ORIGINAL:

Machi [Rename notebook](#) a subset of artificial intelligence that focuses on building systems that learn

SUMMARY:

Machine learning is a subset of artificial intelligence that focuses on building systems that learn
=====

Your max_length is set to 80, but your input_length is only 77. Since this is a summarization task
ORIGINAL:

In recent years, social media has become an essential part of daily communication. Platforms like

SUMMARY:

Social media has become an essential part of daily communication. Many platforms are introducing
=====

ORIGINAL:

The history of space exploration has been marked by significant achievements. From the first human

SUMMARY:

Private companies are entering the space industry, aiming to reduce costs and increase accessibility
=====

```
text = texts[0]
```

```
short_summary = summarizer(text, max_length=40, min_length=20, do_sample=False)
medium_summary = summarizer(text, max_length=80, min_length=40, do_sample=False)
long_summary = summarizer(text, max_length=120, min_length=60, do_sample=False)
```

```
print("SHORT:\n", short_summary[0]['summary_text'])
print("\nMEDIUM:\n", medium_summary[0]['summary_text'])
print("\nLONG:\n", long_summary[0]['summary_text'])
```

Your max_length is set to 120, but your input_length is only 106. Since this is a summarization task
SHORT:

Artificial Intelligence (AI) is rapidly transforming the modern world. It is used in healthcare to

MEDIUM:

Artificial Intelligence (AI) is rapidly transforming the modern world. It is used in healthcare to

LONG:

Artificial Intelligence (AI) is rapidly transforming the modern world. It is used in healthcare to

```
long_text = """Artificial intelligence is transforming industries worldwide. It is used in health
summary = summarizer(long_text, max_length=100, min_length=40, do_sample=False)

print(summary[0]['summary_text'])
```

Your max_length is set to 100, but your input_length is only 94. Since this is a summarization task
Artificial intelligence is transforming industries worldwide. It is used in healthcare, education,

Start coding or [generate](#) with AI.